# 06 Random Forest NoDays

EP

2026-07-22

## Random Forest : Classification

### This notebook mirrors the previous notebook (05__Random__Forest__Classification.Rmd) but drops the predictor `n_days`

For comparison with current classification literature, below I run the richness and Simpson random forest models using classification over regression. The richness and Simpson diversity indices are split based on median, creating two diversity labels: "high" and "low". This mirrors the approach taken by recent studies in which two groups of samples are matched, e.g. 500 "flood" and 500 "no flood" samples.

```
library(targets)
library(tidymodels)
library(ranger)
library(sf)
library(dplyr)

set.seed(399)
```

### Load Data

Read in the final grid model and transform the simpson scores. This isn't strictly necessary for Random Forest but allows for direct comparison with the beta regression model.

```
# Uncomment if using pipeline data:

#setwd("~/CSCT_Pipeline")

#grid_model_all <- tar_read(grid_model_all)

#df <- grid_model_all %>%
  #sf::st_drop_geometry()
```

```
#If running with downloaded grid data use:

grid_model_all <- st_read("Data/grid_model.gpkg", quiet = TRUE)

df <- grid_model_all %>%
  sf::st_drop_geometry()
```

```r
# Adjust Simpson scores
n <- nrow(df)
df$simpson_adj <- (df$simpson * (n - 1) + 0.5) / n

# Set predictors
predictors <- c("dtm_elev", "avg_imperv", "rofsw_risk", "rofrs_risk",
                "avg_area_ha", "treeht", "weighted_precip", "weighted_temp",
                "dist_to_river", "pct_greenspace")
```

### Classification and Fold Setup

```r
#Classification setup
df <- df %>%
  mutate(
    richness_class = factor(
      if_else(sp_richness >= median(sp_richness, na.rm = TRUE), "high", "low"),
      levels = c("low", "high")),
    simpson_class = factor(
      if_else(simpson_adj >= median(simpson_adj, na.rm = TRUE), "high", "low"),
      levels = c("low", "high")))

#Folds & fold city names
folds <- group_vfold_cv(df, group = city)

fold_city_lookup <- folds %>%
  mutate(city = purrr::map_chr(
    splits,~ as.character(unique(assessment(.x)$city)))) %>%
  select(id, city)
```

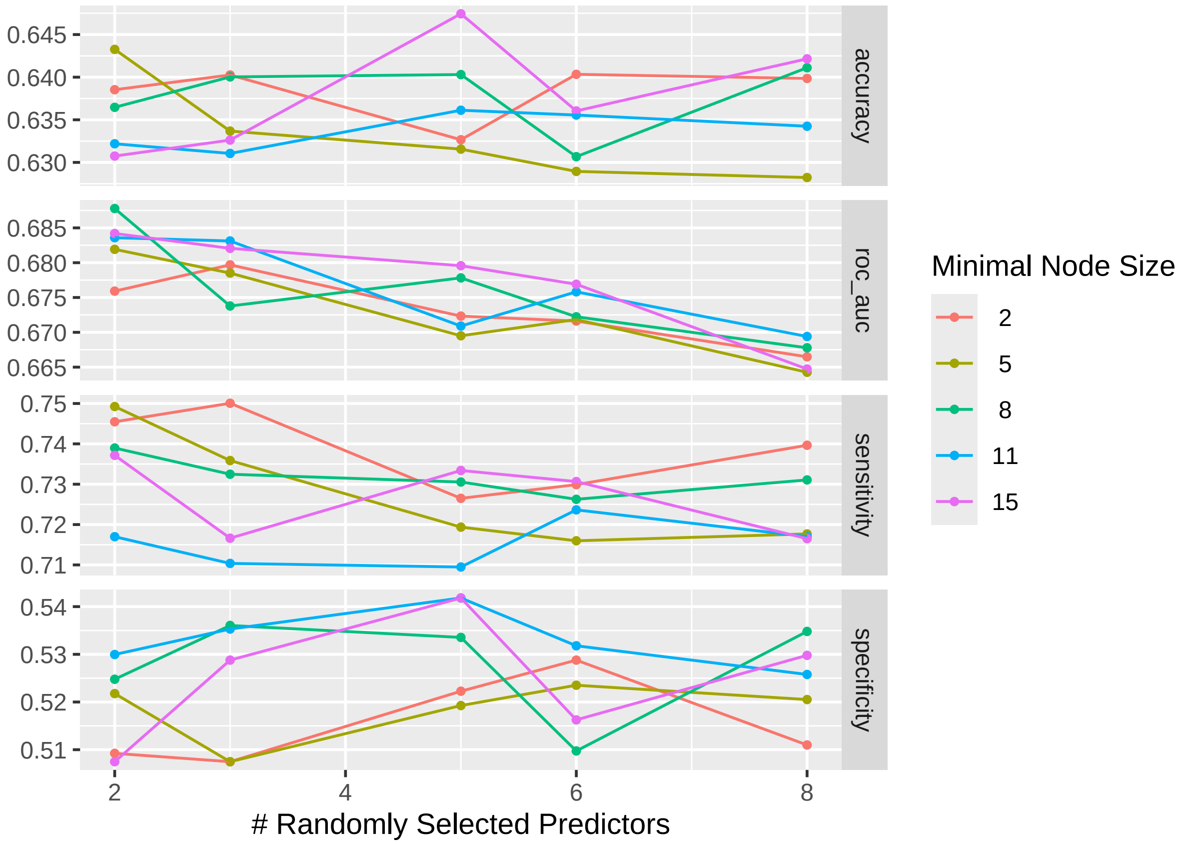
### Model Specification and Recipe

```r
#RF Specification
rf_spec_class <- rand_forest(
  mtry  = tune(),
  min_n = tune(),
  trees = 500) %>%
  set_engine("ranger", importance = "permutation",
             respect.unordered.factors = "order", probability = TRUE) %>%
  set_mode("classification")

#RF Recipe and Workflow
richness_recipe_class <- recipe(
  as.formula(paste("richness_class ~", paste(predictors, collapse = " + "))),
  data = df) %>%
  step_naomit(all_predictors(), all_outcomes())

richness_wf_class <- workflow() %>%
  add_recipe(richness_recipe_class) %>%
  add_model(rf_spec_class)
```

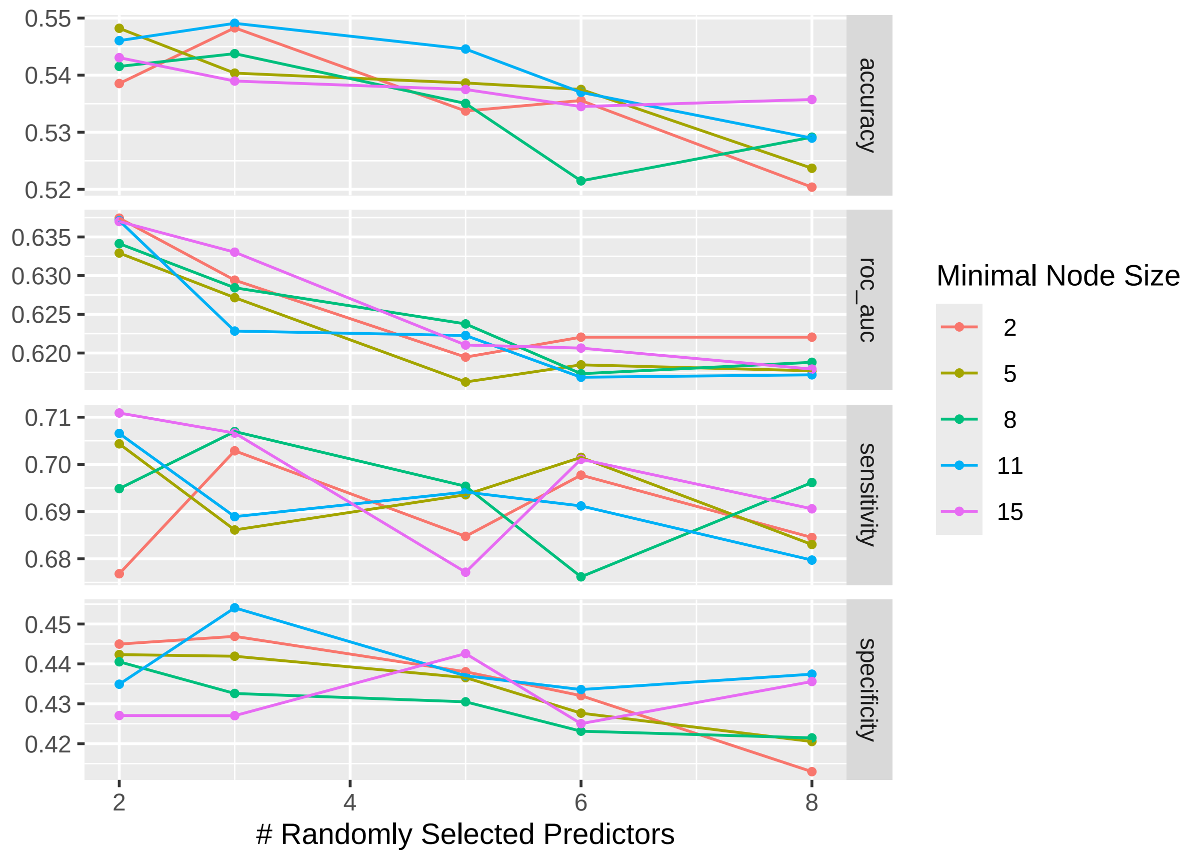
```r
simpson_recipe_class <- recipe(
  as.formula(paste("simpson_class ~", paste(predictors, collapse = " + "))),
  data = df) %>%
  step_naomit(all_predictors(), all_outcomes())

simpson_wf_class <- workflow() %>%
  add_recipe(simpson_recipe_class) %>%
  add_model(rf_spec_class)

rf_grid <- grid_regular(
  mtry(range = c(2, 8)),
  min_n(range = c(2, 15)),
  levels = 5) #Total of 25 combinations in the grid.
```

### Hyperparameter Tuning

```r
richness_res_class <- tune_grid(
  richness_wf_class,
  resamples = folds,
  grid      = rf_grid,
  metrics   = metric_set(roc_auc, accuracy, sensitivity, specificity))

simpson_res_class <- tune_grid(
  simpson_wf_class,
  resamples = folds,
  grid      = rf_grid,
  metrics   = metric_set(roc_auc, accuracy, sensitivity, specificity))

autoplot(richness_res_class)
```




```
autoplot(simpson_res_class)
```




#### Select Best Hyperparameters

```r
show_best(richness_res_class, metric = "roc_auc", n = 5) %>%
  select(mtry, min_n, mean, std_err) %>%
  mutate(across(c(mean, std_err), ~ round(.x, 3))) %>%
  rename("Mean ROC AUC" = mean, "Std. error" = std_err) %>%
  knitr::kable(
    caption = "Top 5 hyperparameter combinations:
    species richness class (ranked by ROC AUC)")
```

Table 1: Top 5 hyperparameter combinations: species richness class (ranked by ROC AUC)

| mtry | min_n | Mean ROC AUC | Std. error |
|---:|---:|---:|---:|
| 2 | 8 | 0.688 | 0.027 |
| 2 | 15 | 0.684 | 0.024 |
| 2 | 11 | 0.684 | 0.025 |
| 3 | 11 | 0.683 | 0.020 |
| 3 | 15 | 0.682 | 0.020 |

```r
show_best(simpson_res_class, metric = "roc_auc", n = 5) %>%
  select(mtry, min_n, mean, std_err) %>%
  mutate(across(c(mean, std_err), ~ round(.x, 3))) %>%
  rename("Mean ROC AUC" = mean, "Std. error" = std_err) %>%
  knitr::kable(
    caption = "Top 5 hyperparameter combinations:
    Simpson diversity class (ranked by ROC AUC)")
```

Table 2: Top 5 hyperparameter combinations: Simpson diversity class (ranked by ROC AUC)

| mtry | min_n | Mean ROC AUC | Std. error |
|---|---|---|---|
| 2 | 2 | 0.637 | 0.015 |
| 2 | 11 | 0.637 | 0.013 |
| 2 | 15 | 0.637 | 0.014 |
| 2 | 8 | 0.634 | 0.014 |
| 3 | 15 | 0.633 | 0.018 |

```r
best_richness_class <- select_best(richness_res_class, metric = "roc_auc")
best_simpson_class  <- select_best(simpson_res_class,  metric = "roc_auc")
```

Compared to the previous notebook containing the full model, the AUC scores are considerably lower for both richness and Simpson models. This is to be expected, since in the regression models we saw a decrease in $R^2$ of 0.606 to 0.302 when removing `n_days`. However, performance is stable across the grid for both models suggesting little room for further tuning.

### Prediction Rate AUC

This looks at the mean AUC across held-out city folds, and individual fold's AUC scores.

```r
# Select best tuning parameters
best_richness_class <- select_best(richness_res_class, metric = "roc_auc")
best_simpson_class <- select_best(simpson_res_class, metric = "roc_auc")

# Overall mean ROC AUC
overall_richness_metrics <- collect_metrics(richness_res_class) %>%
  filter(mtry == best_richness_class$mtry,
         min_n == best_richness_class$min_n,
         .metric == "roc_auc")

overall_simpson_metrics <- collect_metrics(simpson_res_class) %>%
  filter(mtry == best_simpson_class$mtry,
         min_n == best_simpson_class$min_n,
         .metric == "roc_auc")

overall_richness <- overall_richness_metrics$mean
overall_simpson <- overall_simpson_metrics$mean

# ROC AUC by fold (city)
prediction_rate_richness_byfold <- collect_metrics(
```

```r
  richness_res_class, summarize = FALSE) %>%
  filter(mtry == best_richness_class$mtry,
         min_n == best_richness_class$min_n,
         .metric == "roc_auc")

prediction_rate_simpson_byfold <- collect_metrics(
  simpson_res_class, summarize = FALSE) %>%
  filter(mtry == best_simpson_class$mtry,
         min_n == best_simpson_class$min_n,
         .metric == "roc_auc")

# Add city names
city_richness <- prediction_rate_richness_byfold %>%
  left_join(fold_city_lookup, by = "id") %>%
  select(city, .estimate) %>%
  rename(richness_auc = .estimate)

city_simpson <- prediction_rate_simpson_byfold %>%
  left_join(fold_city_lookup, by = "id") %>%
  select(city, .estimate) %>%
  rename(simpson_auc = .estimate)

# Combine city-level results
city_auc <- full_join(city_richness, city_simpson, by = "city")

# Add overall ROC AUC
overall_auc <- tibble(city = "Overall",
                      richness_auc = overall_richness,
                      simpson_auc = overall_simpson)

classification_auc_table <- bind_rows(overall_auc, city_auc) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  rename("City (Held-Out)" = city,
         "Richness ROC AUC" = richness_auc,
         "Simpson ROC AUC" = simpson_auc)

classification_auc_table %>%
  knitr::kable(
    caption = "Overall and city-level cross-validated ROC AUC for species
    richness and Simpson diversity")
```

Table 3: Overall and city-level cross-validated ROC AUC for species richness and Simpson diversity

| City (Held-Out) | Richness ROC AUC | Simpson ROC AUC |
|---|---|---|
| Overall | 0.688 | 0.637 |
| Nottingham | 0.623 | 0.644 |
| Exeter | 0.669 | 0.668 |
| Newcastle | 0.673 | 0.584 |
| Bristol | 0.786 | 0.666 |
| Gateshead | 0.688 | 0.625 |

```r
# Check for any folds where AUC couldn't be computed
richness_na_folds <- prediction_rate_richness_byfold %>%
  filter(is.na(.estimate))

simpson_na_folds <- prediction_rate_simpson_byfold %>%
  filter(is.na(.estimate))

richness_na_folds
```

```
## # A tibble: 0 x 7
## # i 7 variables: id <chr>, mtry <int>, min_n <int>, .metric <chr>,
## #   .estimator <chr>, .estimate <dbl>, .config <chr>
```

```r
simpson_na_folds
```

```
## # A tibble: 0 x 7
## # i 7 variables: id <chr>, mtry <int>, min_n <int>, .metric <chr>,
## #   .estimator <chr>, .estimate <dbl>, .config <chr>
```

There is inter-city variability in AUC score for both species richness (0.786-0.623) and Simpson diversity (0.668 to 0.584) models. Both models have more variation than their full models (previous notebook), suggesting neither generalise to unseen cities as well as their full models.

### Final Model

This is fit on the entire dataset.

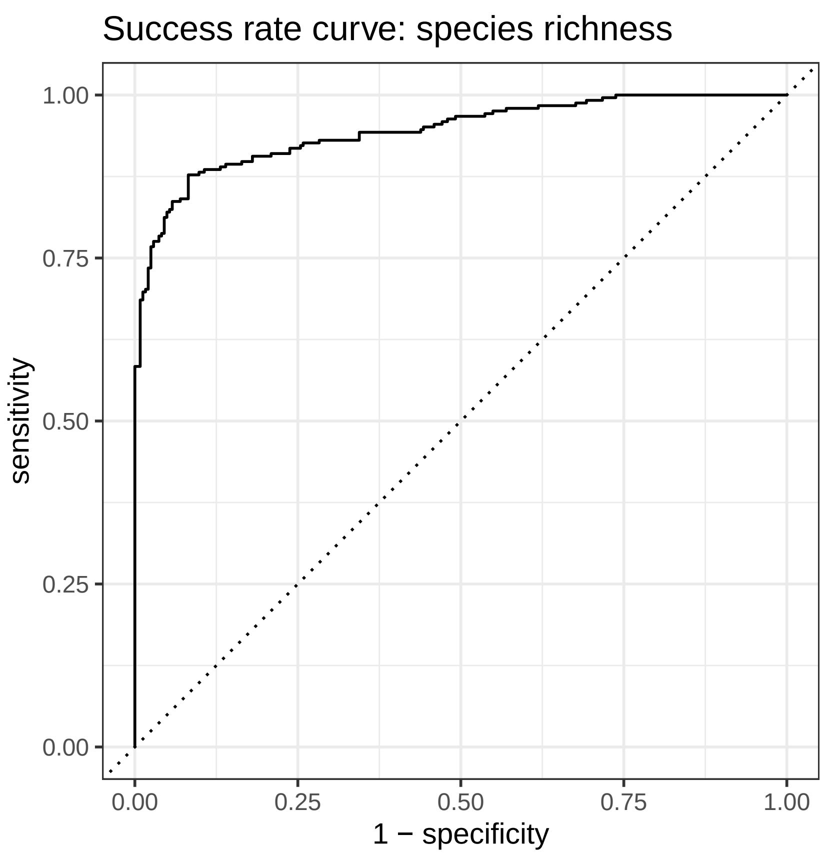
```r
final_richness_wf_class <- finalize_workflow(
  richness_wf_class, best_richness_class)
richness_fit_class <- fit(final_richness_wf_class, data = df)

final_simpson_wf_class <- finalize_workflow(
  simpson_wf_class, best_simpson_class)
simpson_fit_class <- fit(final_simpson_wf_class, data = df)
```

### Success Rate AUC

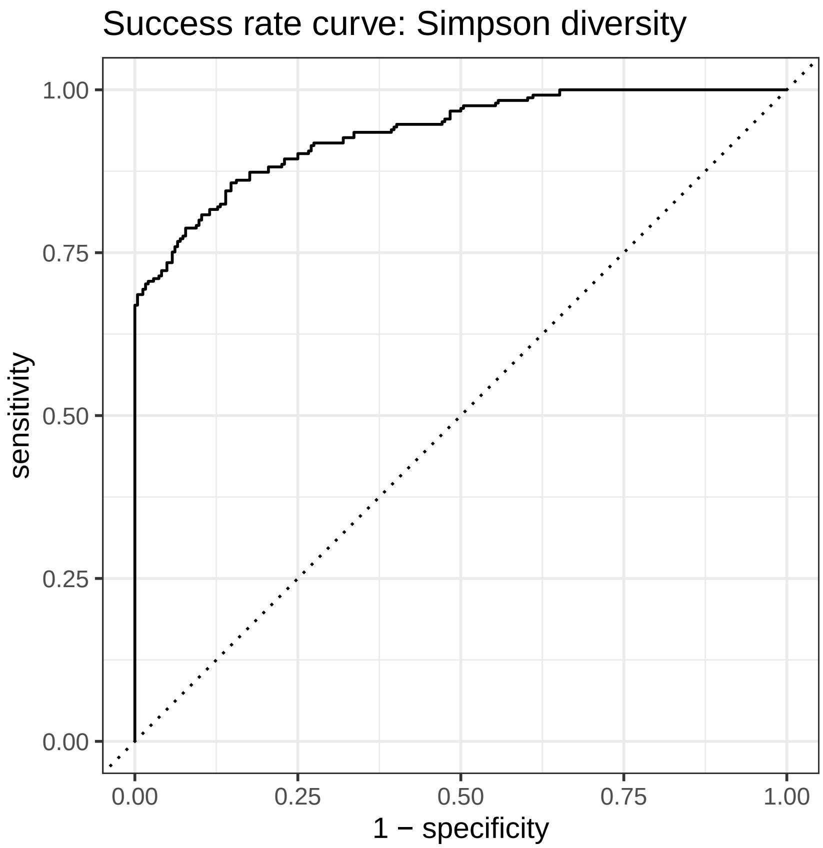
```r
richness_success <- augment(richness_fit_class, new_data = df)
simpson_success  <- augment(simpson_fit_class,  new_data = df)

success_rate_richness <- roc_auc(richness_success, truth = richness_class,
                                 .pred_high, event_level = "second")

success_rate_simpson <- roc_auc(simpson_success, truth = simpson_class,
                                .pred_high, event_level = "second")

#saveRDS(richness_fit_class, "richness_fit_class.rds")
```

**ROC Curves**

```
p1 <- roc_curve(richness_success,
          truth = richness_class,
          .pred_high,
          event_level = "second") %>%
  autoplot() + labs(title = "Success rate curve: species richness")

p2 <- roc_curve(simpson_success,
          truth = simpson_class,
          .pred_high,
          event_level = "second") %>%
  autoplot() + labs(title = "Success rate curve: Simpson diversity")
p1
```




```
p2
```




```r
#ggsave("RN_ND_Rich_Curve.png", p1, width = 12, height = 5, dpi = 300)
#ggsave("RN_ND_Simp_Curve.png", p2, width = 12, height = 5, dpi = 300)
```

**Summary Table : Success Rate vs Prediction Rate AUC**

```r
tibble::tibble(
  Outcome = c("Species richness", "Simpson diversity"),
  `Success rate AUC` = c(success_rate_richness$.estimate,
                         success_rate_simpson$.estimate),
  `Prediction rate AUC` = c(overall_richness,
                            overall_simpson),
  `Prediction rate SE` = c(overall_richness_metrics$std_err,
                           overall_simpson_metrics$std_err)
) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "Random forest classification performance:
             success rate vs. prediction rate AUC")
```

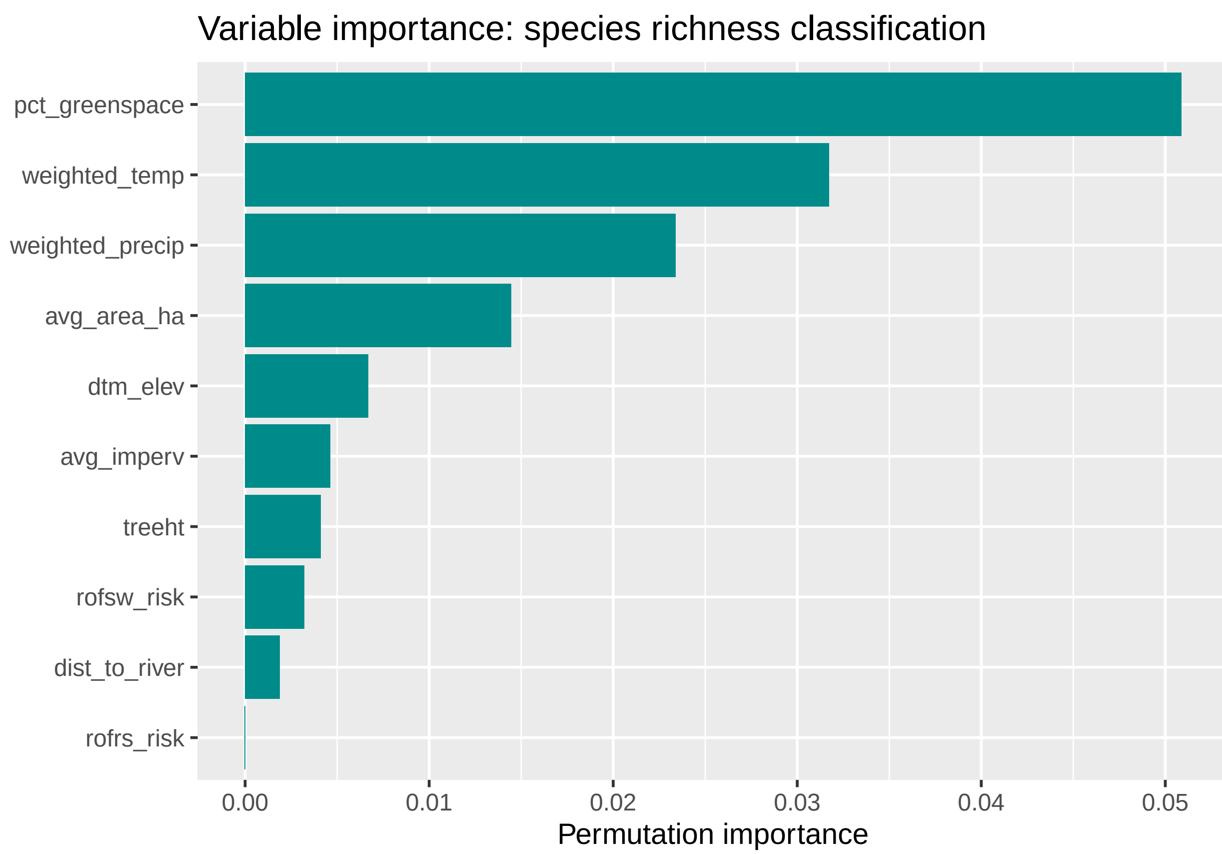
Table 4: Random forest classification performance: success rate vs. prediction rate AUC

| Outcome | Success rate AUC | Prediction rate AUC | Prediction rate SE |
|---|---|---|---|
| Species richness | 0.945 | 0.688 | 0.027 |
| Simpson diversity | 0.934 | 0.637 | 0.015 |

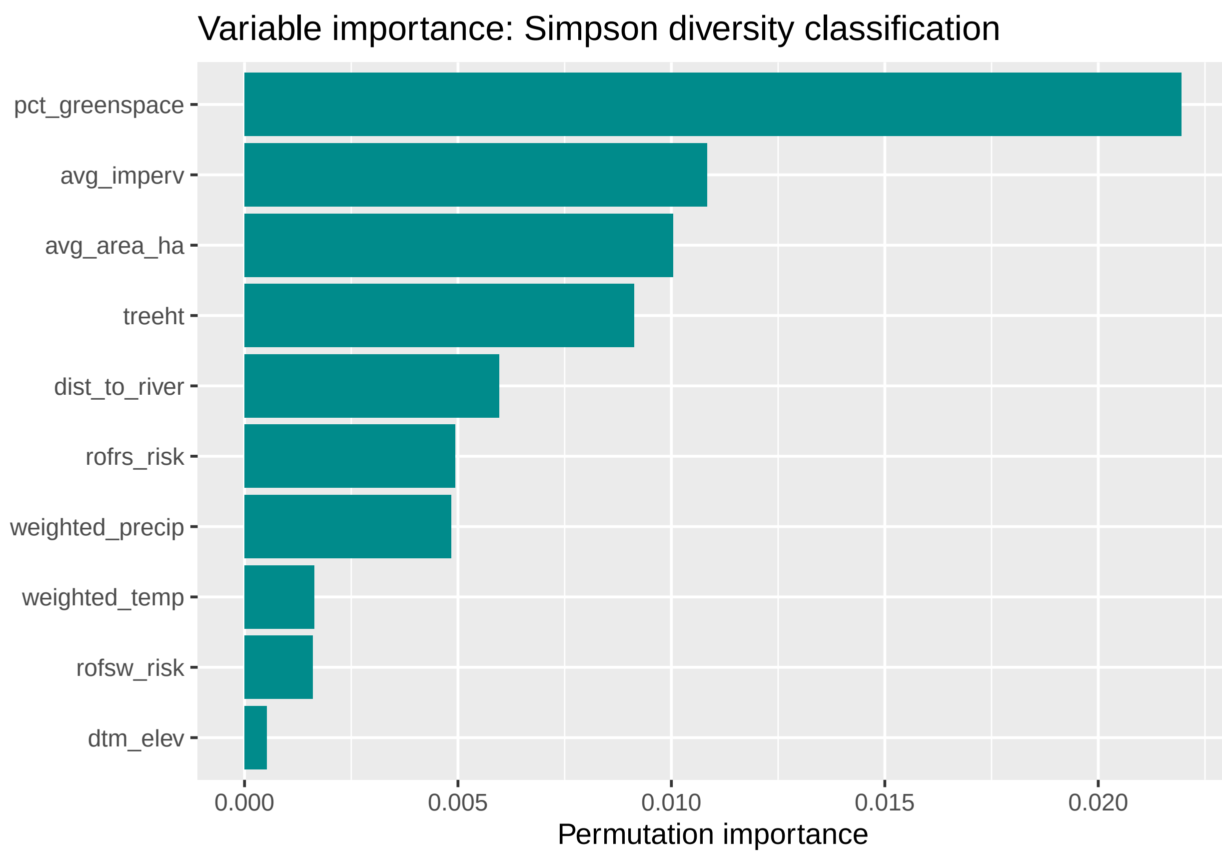
For both models, there is a big decrease in AUC between success rate AUC and prediction rate AUC suggesting that the models fit the training data well but struggle to generalise to unseen cities.

### Variable Importance

```
p3 <- richness_fit_class %>%
  extract_fit_engine() %>%
  ranger::importance() %>%
  sort(decreasing = TRUE) %>%
  as.data.frame() %>%
  tibble::rownames_to_column("variable") %>%
  rename(importance = 2) %>%
  ggplot(aes(x = reorder(variable, importance), y = importance)) +
  geom_col(fill = "cyan4") +
  coord_flip() +
  labs(title = "Variable importance: species richness classification",
       x = NULL, y = "Permutation importance")

p4 <- simpson_fit_class %>%
  extract_fit_engine() %>%
  ranger::importance() %>%
  sort(decreasing = TRUE) %>%
  as.data.frame() %>%
  tibble::rownames_to_column("variable") %>%
  rename(importance = 2) %>%
  ggplot(aes(x = reorder(variable, importance), y = importance)) +
  geom_col(fill = "cyan4") +
  coord_flip() +
  labs(title = "Variable importance: Simpson diversity classification",
       x = NULL, y = "Permutation importance")

p3
```


Variable importance: species richness classification
pct_greenspace
weighted_temp
weighted_precip
avg_area_ha
dtm_elev
avg_imperv
treeht
rofsw_risk
dist_to_river
rofrs_risk
0.00
0.01
0.02
0.03
0.04
0.05
Permutation importance

```
p4
```




```r
#ggsave("VI_RF_ND_Rich.png", p3, width = 12, height = 5, dpi = 300)
#ggsave("VI_RF_ND_Simp.png", p4, width = 12, height = 5, dpi = 300)
```

Despite lower model evaluation scores, the variable importance of predictors is fairly similar to the full model. `pct_greenspace`, `weighted_temp`, and `avg_area_ha` still place highly in the richness model. For the Simpson model, there are similar results again with `pct_greenspace` and `avg_imperv` placing highest.